



OPERATION Department
SIMHADRI

Ref: SMTTP/OPERATION/2022-23/Oprn/167591

Date:

08/06/2022(18:45:33)

Subject: LMI on Guidelines for housekeeping in various areas of coal-based stations

LMI NO: LMI/OGN/OPS/SYST/054

"Guidelines for housekeeping in various areas of coal-based stations"

Prepared on the basis of "COS-ISO-00-/OGN/OPS/SYST/054"

Put up for kind approval.

Submitted by:

जुलियट जॉनी - इंजीनियर

JULIET JOHNY - 105435 (ENGINEER ,OPERATION) ,Simhadri
(JULIETJOHNY@NTPC.CO.IN, 7597808553)

On: 08/06/2022(18:45:33) **Ref:** SMTTP/OPERATION/2022-23/Oprn/167591

Submitted by:

टी वी राजशेखर - अपर महाप्रबंधक

T V Rajasekhar - 007005 (ADDL. GENERAL MANAGER ,OPERATION)
(TVRSEKHAR@NTPC.CO.IN, 9650998176) ,Simhadri

On: 08/06/2022(18:46:58) **Ref:** SMTTP/OPERATION/2022-23/Oprn/167591

May pl be approved

Submitted by:

राजशेखर पाला - अपर महाप्रबंधक

Rajasekhar Pala - 005146 (ADDL. GENERAL MANAGER ,OPN (MAIN PLANT))
(PRAJSEKHAR@NTPC.CO.IN, 8331015741) ,Simhadri

On:08/06/2022(19:39:17) **Ref:**SMTTP/OPERATION/2022-23/Oprn/167591

Forwarded for further process please.

Submitted by:

शंभू कुमार सुमन - अपर महाप्रबंधक

Shambhu Kumar Suman - 005007 (ADDL. GENERAL MANAGER ,MECH MAINT- TURBINE)
(SKSUMAN@NTPC.CO.IN, 9431215296) ,Simhadri

On:09/06/2022(10:33:50) **Ref:**SMTTP/OPERATION/2022-23/Oprn/167591

Submitted for approval

Submitted by:

ओम प्रकाश - अपर महाप्रबंधक

Om Prakash - 007105 (ADDL. GENERAL MANAGER ,MECH MAINT- BOILER)
(OMPRAKASH08@NTPC.CO.IN, 9650994070) ,Simhadri

On:10/06/2022(18:24:27) **Ref:**SMTTP/OPERATION/2022-23/Oprn/167591

Submitted by:

नीलाजन मंडल - अपर महाप्रबंधक

Nilanjan Mandal - 007418 (ADDL. GENERAL MANAGER ,ASH HANDLING)
(NILANJANMANDAL@NTPC.CO.IN, 9434097418) ,Simhadri

On:10/06/2022(19:01:35) **Ref:**SMTTP/OPERATION/2022-23/Oprn/167591

Submitted by:

धर्मेन्द्र कुमार तोमर - अपर महाप्रबंधक

Dharmendra Kumar Tomar - 007286 (ADDL. GENERAL MANAGER ,ELECT MAINT)
(DHARMENDRAKUMAR02@NTPC.CO.IN, 9416212079) ,Simhadri

On:10/06/2022(19:51:53) **Ref:**SMTTP/OPERATION/2022-23/Oprn/167591

Housekeeping points related to Swgrs, Cable galleries covered and complied.

LMI forwarded for approval

Submitted by:

रवि वर्मा कनुमुरी - उप महाप्रबंधक

Ravivarma Kanumuri - 008001 (DY. GENERAL MANAGER ,ELECT MAINT- MAIN PLANT)
(RAVIVARMAK@NTPC.CO.IN, 9493174808) ,Simhadri

On:11/06/2022(09:17:13) **Ref:**SMTPP/OPERATION/2022-23/Oprn/167591

Submitted by:

धर्मेन्द्र कुमार तोमर - अपर महाप्रबंधक

Dharmendra Kumar Tomar - 007286 (ADDL. GENERAL MANAGER ,ELECT MAINT)
(DHARMENDRAKUMAR02@NTPC.CO.IN, 9416212079) ,Simhadri

On:11/06/2022(09:37:01) **Ref:**SMTPP/OPERATION/2022-23/Oprn/167591

LMI forwarded for approval.

Submitted by:

आन्नप्रगद रमा कोटेश्वरराव - अपर महाप्रबंधक

Annapragada Rama Koteswararao - 047211 (ADDL. GENERAL MANAGER ,FUEL HAND.- OPN)
(ARKOTESWARARAO@NTPC.CO.IN, 8332988907) ,Simhadri

On:13/06/2022(16:17:56) **Ref:**SMTTP/OPERATION/2022-23/Oprn/167591

May review with respect to OGN issued by COS, as there is no coverage of CERs, UCRs, Server Rooms, 8.5m UPS area etc., in this LMI

Submitted by:

टी.आर. राजगोपालन् - अपर महाप्रबंधक

TR Rajagopalan - 003994 (ADDL. GENERAL MANAGER ,C & I MAINT)
(TRRAJAGOPAL@NTPC.CO.IN, 9440918024) ,Simhadri

On:13/06/2022(17:33:05) **Ref:**SMTTP/OPERATION/2022-23/Oprn/167591

- House keeping of **CER and associated server rooms** at TG floor and **DCS and UPS rooms at 8.5 mtr. elevation** are very important along with control room.

- Neccesary corrections may be done to include the above areas at **sr. no.4.6.3** .

Submitted by:

चौ. किशोर कुमार - अपर महाप्रबंधक

Ch Kishore Kumar - 006106 (ADDL. GENERAL MANAGER ,C & I MAINT)
(CHKISHOREKUMAR@NTPC.CO.IN, 9440918196) ,Simhadri

On:14/06/2022(10:40:17) **Ref:**SMTTP/OPERATION/2022-23/Oprn/167591

The areas mentioned in Pg 6 may be included in the scope of the LMI and put up for further approval. In the Section 5.0 (Records to be maintained), certification for the above areas will be done by C&I shift engineer on daily basis

Submitted by:

टी.आर. राजगोपालन् - अपर महाप्रबंधक

TR Rajagopalan - 003994 (ADDL. GENERAL MANAGER ,C & I MAINT)
(TRRAJAGOPAL@NTPC.CO.IN, 9440918024) ,Simhadri

On:14/06/2022(10:53:52) **Ref:**SMTTP/OPERATION/2022-23/Oprn/167591

Pl incorporate the comments of C&I dept in the LMI.

Submitted by:

राजशेखर पाला - अपर महाप्रबंधक

Rajasekhar Pala - 005146 (ADDL. GENERAL MANAGER ,OPN (MAIN PLANT))
(PRAJSEKHAR@NTPC.CO.IN, 8331015741) ,Simhadri

On:15/06/2022(19:37:19) **Ref:**SMTTP/OPERATION/2022-23/Oprn/167591

Comments from C&I Department have been added in the LMI

Submitted by:

जुलियट जॉनी - इंजीनियर

JULIET JOHNY - 105435 (ENGINEER ,OPERATION)
(JULIETJOHNY@NTPC.CO.IN, 7597808553) ,Simhadri

On:16/06/2022(17:56:48) **Ref:**SMTTP/OPERATION/2022-23/Oprn/167591

C&I remarks have been incorporated in LMI.

Submitted by:

राजशेखर पाला - अपर महाप्रबंधक

Rajasekhar Pala - 005146 (ADDL. GENERAL MANAGER ,OPN (MAIN PLANT))
(PRAJSEKHAR@NTPC.CO.IN, 8331015741) ,Simhadri

On:18/06/2022(15:40:29) **Ref:**SMTTP/OPERATION/2022-23/Oprn/167591

Submitted by:

Vishwa Mohan Choudhary - 003932 (GENERAL MANAGER ,FUEL MANAGEMENT)
(VMCHOUDHURY@NTPC.CO.IN, 7588691232) ,Simhadri

On:18/06/2022(15:54:25) **Ref:**SMTTP/OPERATION/2022-23/Oprn/167591

Submitted for approval of competent authority

Submitted by:

झिमलि धर - अपर महाप्रबंधक

Jhimli Dhar - 006460 (ADDL. GENERAL MANAGER ,EEMG)
(JHIMLIDHAR@NTPC.CO.IN, 9650998085) ,Simhadri

On:21/06/2022(15:43:14) **Ref:**SMTTP/OPERATION/2022-23/Oprn/167591

May please be accorded approval.

Submitted by:

मैथ्यू ई कोवूर - महाप्रबंधक

Mathew Eipe Kovoor - 005637 (GENERAL MANAGER ,MAINTENANCE)
(MEKOVOOR@NTPC.CO.IN, 9650994652) ,Simhadri

On:25/06/2022(11:45:27) **Ref:**SMTTP/OPERATION/2022-23/Oprn/167591

Submitted by:

Dhurjati Prasad Patra - 004573 (GENERAL MANAGER ,OPERATION)
(DPPATRA@NTPC.CO.IN, 8275045446) ,Simhadri

On:25/06/2022(18:34:34) **Ref:**SMTTP/OPERATION/2022-23/Oprn/167591

समीक्षित। Reviewed.

कृपया अनुमोदन प्रदान किया जाए। May be approved please.

Submitted by:

राजेश कुमार झा - महाप्रबंधक

Rajesh Kumar Jha - 003323 (GENERAL MANAGER ,O & M)
(RKJHA02@NTPC.CO.IN, 9415342046) ,Simhadri

On:25/06/2022(20:43:24) **Ref:**SMTTP/OPERATION/2022-23/Oprn/167591

Approved & Submitted by:

गिरीश चन्द्र चौकसे - महाप्रबंधक

Girish Chandra Choukse - 002907 (GENERAL MANAGER ,HEAD OF PROJECT)
(GCCHOUKSE@NTPC.CO.IN, 9425178006) ,Simhadri

On:27/06/2022(10:20:31) **Ref:**SMTTP/OPERATION/2022-23/Oprn/167591

NTPC Ltd.

Simhadri Thermal Power Station

LOCATION MANAGEMENT INSTRUCTION

LMI/OGN/OPS/SYST/054

Issue No.: 1 Rev: 0 Date: JUNE 2022

**Guidelines for housekeeping in various areas of coal-based
stations**

**Approved for
Implementation by
HOP (Simhadri)**



Q & M - OPERATION DEPARTMENT

Ref: SMPP/OPRN/LMI/OGN/OPS/SYST/054

Date: 07.06.2022

The following LMI has been revised in line with revised OD vide “COS-ISO-00-/OGN/OPS/SYST/054, Rev no:0, Date: MAY’ 2022”

Sl.No	LMI revision details	LMI (Name)	New LMI	LMI Created Dt.
1	LMI/OGN/OPS/SYST/054 Issue no: 1 Rev no:0, Dt: MAY’2022	Guidelines for housekeeping in various areas of coal- based stations	OGN by CC- OS	MAY 2022

Put up for kind approval please

	Name	Designation	Signature
Prepared by:	JULIET JOHNY	Engineer (O)	
Checked by	T V RAJASEKHAR	AGM(O)	
Re Checked by:	PALA RAJASEKHAR	AGM(O)	
Checked by	S K SUMAN	AGM(TMD)	
Checked by	OM PRAKASH	AGM(BMD)	
Checked by	Nilanjan Mandal	AGM(AHP)	
Checked by	D.K Tomar	AGM(EMD)	
Checked by	A R Koteswara RAO	AGM(CHP)	
Checked by:	T R RAJAGOPALAN	AGM(C&I)	
Rechecked by	V M Choudhary	GM(FM)	
Rechecked by	MATHEW EIPE KOVOOR	GM(Maint)	
Reviewed by:	R K JHA	GM (O&M)	
Approved by	G C CHOUKSE	GGM (Simhadri)	

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1.0 SUMMARY

1.1.	Superseded Documents	Not Applicable
1.2.	Scope	
	This LMI is applicable to all four 500MW units of Simhadri STPS (Stage-I&II) their auxiliaries and offsite pump houses and CHP.	
1.3	Review	
	Review Responsibility	GM (Operation)
	Review Frequency	2 Year basis or as and when require, whichever is lower
	Review Approval	Head of operation

2.0 INTRODUCTION

- a. Good housekeeping involves every phase of industrial operations and should apply throughout the entire premises, indoors and out. It is more than mere cleanliness. It requires orderly conditions, the avoidance of congestion, and attention to such details as an orderly layout of the whole workplace, the marking of aisles, display boards for safe housekeeping activities, Display of Hazards on site, adequate storage arrangements, and suitable provision for cleaning and maintenance.
- b. Good housekeeping helps to create:
 - *Better working conditions*
 - *Safer workplaces*
 - *Greater efficiency.*
- c. Poor housekeeping can also lead to infestation by pests such as rodents and cockroaches and create serious health risks.
- d. These guidelines are emphasized towards safe housekeeping methods for various areas of thermal power plant and handling various types of dust and dirt present in that location.

3.0 RECORDS OF REVISION

Not Applicable

4.0 PROCESS

4.1. GENERAL DESCRIPTION

- Good housekeeping is a vital factor in preventing accidents. The great majority of all work accidents are caused during the handling of goods or materials, and by people falling, being hit by falling objects, or striking against objects in the workplace. All these causes can be reduced by good housekeeping practices—in fact, good housekeeping is the only cure for hundreds of accidents that occur.
- Here are some kinds of accidents commonly caused by bad housekeeping:
 - Tripping over loose objects on floors, stairs and platforms.
 - Articles dropping from above.
 - Slipping on greasy, wet or dirty surfaces.
 - Striking against projecting, poorly stacked, or misplaced material.
 - Cuts on hands or other parts of the body by projecting nails, wire, steel strapping on bales or crates, scrap etc.
- 4.1.1. Where housekeeping is bad, fire is a constant hazard. It can be caused by many housekeeping problems—such as oil-soaked rags and clothing igniting from spontaneous combustion; Accumulated inflammable coal dust which might catch fire: or piles of paper and other packing materials being allowed to accumulate.
- 4.1.2. It is not only floors that need sweeping. Dust and grime also collect on ledges, shelves, piping, conduits, lamps, reflectors, windows, cupboards, lockers, and so on—and all these places need attention.
- 4.1.3. As coal based power plants deals with coal there are many chances of generating coal dust during the process of power generation which needs to be cleaned.
- 4.1.4. Cleaning coal dust poses many challenges because it can cause fire hazard and may cause harm to human beings and nearby equipment. As the volatile matter content in coal increases, the dust from coal becomes more prone to fire and explosion.

4.2. HOUSEKEEPING METHODOLOGY

- 4.2.1. The methods used for cleaning surfaces shall be selected on the basis of reducing the potential for creating a combustible dust cloud.
 - a. Vacuum cleaning method: The Dust is sucked in the dust chamber with the help of vacuum cleaner so as to avoid dust cloud formation.
 - b. Sweeping, shoveling, scoop, and brush cleaning method.
 - c. Water washdown cleaning method:
 - d. Water foam washdown systems.
 - e. Compressed air blowdown methods.
- 4.2.2. Precautions to be taken as per housekeeping methodology adopted.
 - i. Due care should be taken while vacuum cleaning to avoid entry of burning/smoldering coal dust in the dust chamber of vacuum cleaner.
 - ii. Sweeping, shoveling technique can also generate coal dust cloud if done very

- rigorously, same to be avoided.
- iii. Usage of metal scrappers (Belcha) and tools which can generate spark while cleaning coal dust should also be avoided.
 - iv. Water stream should not be used to clean heavy coal dust as it can generate dust clouds.
 - v. While doing any water washing activity, availability of drain path must be ensured.
 - vi. Make sure equipment (vacuum cleaner) is grounded to avoid electrical and static sparking.
 - vii. Only use vacuum cleaners approved for dust collection.
 - viii. Water washing should not be used to smoldering coal/coal dust as it may pose water gas hazard.

Note: Where blowdown using compressed air is used, the following precautions shall be followed:

- A. Prior to using compressed air, vacuum cleaning, sweeping, or water washdown methods are used to clean surfaces that can be safely accessed.
- B. All electrical equipment, including lighting, potentially exposed to airborne dust in the area during cleaning should be of spark proof rating.
- C. All ignition sources and hot surfaces capable of igniting a dust cloud or dust layer are shut down or removed from the area.
- D. After blowdown is complete, residual dust on lower surfaces is cleaned prior to re-introduction of potential ignition sources.

4.3. EVALUATION OF HOUSEKEEPING HAZARDS

- 4.3.1. Slippery surface- Slip and fall hazard
- 4.3.2. Material fall hazard: Loose material over head
- 4.3.3. Work at Height-Fall Hazard
- 4.3.4. Combustible dust:
- 4.3.5. Combustible dust is a major hazard in thermal power plants. Dust may accumulate on almost any surface, including angled or vertical surfaces, depending on the particle size or charge. Accumulations may also occur on surfaces that are not visible without looking beyond the ground level. The exterior and immediate area surrounding process equipment that generates dust, collects dust, and/or transfers materials containing dust (e.g., crushers, grinders, Trippers, feeders, ductwork, pipes, belt conveyors, bucket elevators, screw conveyors, or connectors) are also especially prone to dust accumulation.
- 4.3.6. To be explosive, the dust mixture should have:
 - Particles dispersed in the air with minimum size (typical figure is 400 microns);
 - Dust concentrations must be reasonably uniform; and
 - Minimum explosive concentration for coal dust (33% volatiles) is 50 gm/m³.

- 4.3.7.** Failure of dust extraction and suppression systems may lead to abnormal conditions and may increase the concentration of coal dust to the explosive limits. Sources of ignition present are incandescent bulbs with the glasses of bulkhead fittings missing, electric equipment and cables, friction, spontaneous combustion in accumulated dust.
- 4.3.8.** An initial (primary) explosion (see Figure 1) in processing equipment or in an area where fugitive dust has accumulated may shake loose more accumulated dust, or damage a containment system (such as a duct, vessel, or collector). As a result, if ignited, the additional dust dispersed into the air may cause one or more secondary explosions (see Figure 1). These can be far more destructive than a primary explosion due to the increased quantity and concentration of dispersed combustible dust.

Fig-1 Primary and secondary Explosion

- 4.3.9.** Stockpile areas shall be provided with stockpile dust suppression system as well as to reduce spontaneous ignition of the coal stockpiles. Necessary water distribution network for drinking and service water with pumps, piping, tanks, valves etc will be provided for distributing water at all transfer points, crusher house, control rooms etc. A centralized control room
- 4.3.10.** Controlling the Explosion Pentagon

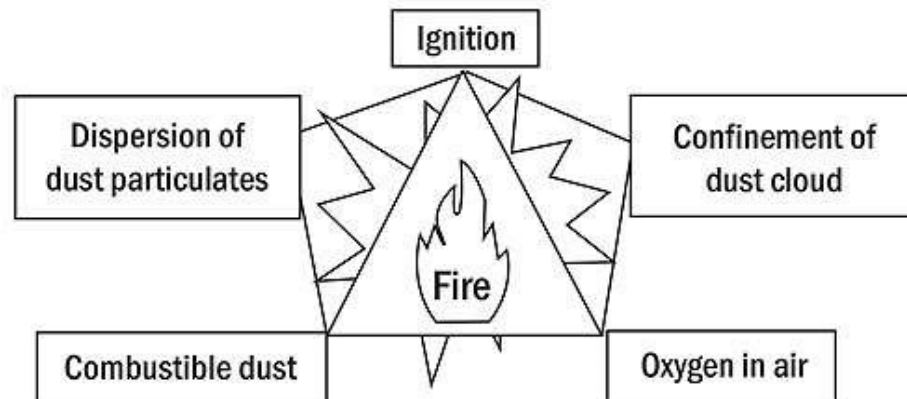
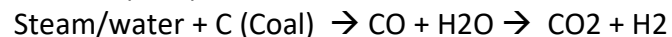
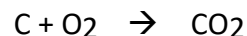
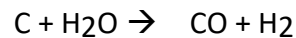


Fig-2 Explosion Pentagon

- 4.3.11.** The explosion pentagon (Fig-2) includes the three elements of the fire triangle: fuel (combustible dust), ignition source (heat) and an oxidizer (air). However, an explosion also needs two additional elements: dispersion of dust particles (in sufficient quantity and concentration) and the confinement of the dust cloud (vessel, area or building).

4.3.12. If one of the elements of the explosion pentagon is missing, a catastrophic explosion cannot occur. Two of the elements in the explosion pentagon are difficult to eliminate: oxygen (within air), and confinement of the dust cloud (within processes or buildings). However, the other three elements of the pentagon can be controlled to a significant extent and will be discussed further in this document.

4.3.13. Hazard due to water Gas: When the water is subjected on red hot coal or smoldering coal dust (large quantity), it becomes very hot water stream and may undergo coal gasification processes. CO, CO₂, H₂, H₂O, CH₄, Sulfur-oxides, etc. may form under these severe conditions. This is nothing but “Water Gas”, a mixture of carbon monoxide and Hydrogen. The generated CO further reacts with steam / water to generate Hydrogen and accelerates the generation of Hydrogen gas. Once the Hydrogen percentage reached 4% (volume by volume) in ambient air, explosion takes place without any external energy, with the release of enormous energy and heavy shock waves.



4.4. Facility Analysis Components and SOP preparation.

4.4.1 Every station must prepare SOPs for housekeeping w.r.t various Areas along with checklists which should carefully identify the following in order to assess their potential for any hazard as mentioned in above sections.

- Materials that can be combustible when finely divided;
- Processes which use, consume, or produce combustible dusts;
- Open areas where combustible dusts may build up;
- Hidden areas where combustible dusts may accumulate.
- Methodology of housekeeping to be adopted.
- list of PPEs required during house keeping
- Precautions to be taken during housekeeping.
- Means by which dust may be dispersed in the air; and
- Potential ignition sources and steps to mitigate them.

4.4.2 Places where dust accumulations are most likely include:

- Floors, railings, structural surfaces, and processing equipment surfaces.
- Overhead structural supports and components, ductwork, pipes, trays, conduit, and pipe racks.

- Open control rooms, storage rooms, and other areas built inside or adjacent to larger production areas where dust has accumulated.
- Above ceilings (e.g., hidden surfaces behind false ceilings, ceiling tiles, or ceiling plenum areas), behind walls, behind wall coverings, and inside ventilation ducts.

4.4.3 Places where Coal dust generates:

1. Boiler Firing Floor
2. Milling area.
3. Feeder Floor area
4. Bunker feeding area.
5. Coal conveyor area
6. Crusher house
7. Coal TP house
8. Coal unloading area
9. Paddle feeder area.
10. Coal stacking area.
11. Any other area identified.
12. Mill Reject area.

4.5. MITIGATION OF HOUSEKEEPING HAZARDS

- 4.5.1. Where coal-handling operations may produce a combustible atmosphere from fuel sources or from flammable gases or dust, sources of ignition shall be eliminated or safely controlled to prevent ignition of the combustible atmosphere.
- 4.5.2. Places where coal dust accumulation takes place should be cleaned by vacuum cleaning method to prevent any coal dust cloud formation.
- 4.5.3. Limit/reduce the amount of overhead horizontal surfaces (racks, piping, ductwork, drop ceiling), which trap dust and are hard to clean.
- 4.5.4. If cleaning is to be done in a closed chamber, a confined space PTW has to be taken and OGN-OPS-SYST-047 "Guidelines for Working in Confined Space" to be followed.
- 4.5.5. If Cleaning is to be done at heights like Conveyor Take ups "Work at Height" PTW must be taken with proper JSA and vertigo test of worker involved. And in such cases one person should be at ZERO meter to warn others moving around about housekeeping being done on height or with barricading.
- 4.5.6. Any Source of leakage or waste material for which housekeeping is done, should be identified at the earliest and WOC must be raised to attend such defects.
- 4.5.7. All identified areas where PF leakages are frequent like, Burner zone, Mill discharge valve area, and mill outlet PF pipes should be inspected with every PM (Preventive maintenance) of mills.

- 4.5.8. Any leakage of air in Burner zone should also be attended on priority as it can cause dust clouds if coal dust accumulation is there.
- 4.5.9. Hot air leakages which can increase the temperature in nearby area should be attended on priority as it can cause fire hazard in nearby area.
- 4.5.10. Any oil leakages from oil tanks or lines should be attended on priority and if because of any reason it could not be attended proper marking should be done with leakage oil collecting mechanism, area should be clear for any combustible material and heat/ignition source.
- 4.5.11. Stations where box type girders are used in boiler, all entry points to box type girders should be kept sealed to prevent any dust ingress inside.
- 4.5.12. Provide access to all hidden areas to permit inspection.
- 4.5.13. Inspect for dust residues in open and hidden areas, at regular intervals.
- 4.5.14. Keep the dust level below 1/32" on horizontal surfaces in all the areas by appropriate scheduling of housekeeping.
- 4.5.15. Dust suppression/collection(if any) system shall be kept in service as per OEM guidelines.
- 4.5.16. All the locations where housekeeping is a regular practice should be surveyed for potential hazards that are present and may arise during the housekeeping process. The identified hazards should be documented with proper housekeeping methods to mitigate the hazard for that area.
- 4.5.17. The personnel deployed for housekeeping should be well informed reg. various hazards associated with housekeeping of that area.
- 4.5.18. All the Necessary PPEs must be worn by housekeeping staff i.e. Gloves, respiratory mask, Goggles, safety shoes etc.. Housekeeping staff deployed in areas which are identified for combustible dust hazard should be equipped with flame resistant clothing along with all PPEs.
- 4.5.19. Housekeeping staff should be trained for carrying out housekeeping jobs in consideration of all the hazards present in the area along with knowledge reg. usage of waste chutes, Operating vacuum cleaners, waste disposal etc.
- 4.5.20. Dust generated from Biomass and pellets pose a serious hazard as their VM content is very high.
- 4.5.21. Extinguishing fire of coal/coal dust: Remove ignition source and "fuel supply" from fire if possible to do so safely. For small fires use dry chemical, sand, earth, water spray (not stream), or regular foam. For a massive fire use unmanned hose holder or monitor nozzles. If this is impossible, withdraw from the area and let the fire burn, concentrating on protecting the surrounding assets from the fire.
- 4.5.22. While cleaning finished floors like tiles, marble etc, a caution board must be used for slip and fall hazard.

4.6. AREA WISE HOUSEKEEPING GUIDELINES:

4.6.1. Boiler Vertical----- Responsibility Operation dept

- 4.6.1.1. Boiler should be cleared for any loose inflammable material like papers, glass wool, oil spills etc.
- 4.6.1.2. There should not be any material obstructing the boiler walkways.
- 4.6.1.3. Areas where coal dust gets accumulated shall be first cleaned by vacuum cleaning method only.
- 4.6.1.4. If even after vacuum cleaning further cleaning is required then brooms should be used to accumulate the dust and vacuum cleaned.
- 4.6.1.5. Usage of water jets and compressed air should be avoided for cleaning coal dust.
- 4.6.1.6. Any horizontal surfaces in girders where coal dust may accumulate should be sealed/covered to avoid any accumulation.

4.6.2. Boiler zero level-----Responsibility Operation dept

- 4.6.2.1. Boiler Zero level can be cleaned by any of the methods except compressed air.
- 4.6.2.2. Before using water jets care should be taken by ensuring water draining path availability.
- 4.6.2.3. Equipment and instruments placed in fields shall be cleaned by compressed air or blower.

4.6.3. TG area including control room, CERs, Server rooms, DCS & UPS Room----- Resp. Operation & C&I dept

- 4.6.3.1. These areas should be cleaned by brooms and mops.
- 4.6.3.2. vacuum cleaning method may also be adopted.
- 4.6.3.3. Cobwebs in above mentioned areas shall be cleaned regularly.
- 4.6.3.4. Equipment and instruments placed in fields shall be cleaned by compressed air or blower.
- 4.6.3.5. Wastepaper baskets / trash bins/scrap material in the above areas shall be cleared every day and transported by the agency to a suitably designated place as and when required.
- 4.6.3.6. Unit control room, C&I rooms, control equipment rooms, UPS rooms, C&I lab and operator rooms shall be wet mopped (wet process) every day.

4.6.4. Electrical panel/switchgear/battery rooms/cable gallery-----Res. EMD

- 4.6.4.1. Cleaning of electrical panel/switchgear/battery rooms/cable galleries shall only be done under proper supervision.
- 4.6.4.2. Loose cloth wear shall not be allowed for housekeeping near any electrical panel/equipment.
- 4.6.4.3. Housekeeping materials shall not be stored in electrical panel/battery rooms.
- 4.6.4.4. All the electrical panel/switchgear/battery rooms/cable galleries are to

- be kept free from cobwebs.
- 4.6.4.5. Dusting of electrical panels shall not be done by general housekeeping workmen.
 - 4.6.4.6. Dusting of only specified electrical equipment/instruments by electrical maintenance department should be included in housekeeping. Electrical maintenance department shall ensure there is no electrical hazard/system reliability issue due to regular maintenance on identified equipment's.
 - 4.6.4.7. It shall be ensured that there is no disinfectant or water spray/spill on electrical panels/equipment during housekeeping.
 - 4.6.4.8. Wet moping shall only be done in switchgear rooms/cable galleries. NO water washing /water spraying should be allowed in any electrical panel/switchgear room/cable gallery during housekeeping.
 - 4.6.4.9. Use of broom should be avoided in electrical panel /switchgear/battery rooms to avoid any dust suspension in air which may have an impact on electrical systems reliability. Use of vacuum cleaners may be encouraged for cleaning of floor.
 - 4.6.4.10. Any sort of debris on floor of cable galleries is to be cleaned regularly. Any debris on cable trays/trenches shall be cleaned only under proper supervision.
 - 4.6.4.11. Cable trays are to be dusted preferably during unit shutdown and using dry compressed air only.
 - 4.6.4.12. Fiberglass/wood ladders shall only be used in electrical panel room/battery room/switchgear room/cable gallery if required. Aluminum ladders shall not be used in any case.

4.6.5. Coal Handling Plant-----Responsibility CHP dept

4.6.5.1. Coal Unloading/Coal evacuation Zone

4.6.5.1.1. Wagon Tippler Complex

- i. Track Area & adjacent pavement should be well illuminated and clear of all coal/material/debris/scrap so as to ensure smooth movement of SAC/Wagons and Man/material respectively.
- ii. Proper Arrangement of Stone handling equipment to be ensured so as to ensure safe handling of boulder/stone pieces by housekeeping staff.
- iii. For Wagon Tippler Grill Cleaning adequate arrangement for Pay loaders/Backhoe excavators should be made with all safety precautions.
- iv. Wagon Tippler Pit Cleaning Should be done on regular basis.
- v. Availability of sump/slurry Pumps shall be ensured for underground portion of wagon tippler complex to avoid water logging/ coal slurry accumulations.
- vi. Apron Feeder Floor must be cleaned on regular basis and should be clear of all coal/slurry/debris/scrap
- vii. The Drive Area/ Return side of Apron Feeder & dribble Conveyor should

be clean. Proper Covers/Guards should be in place for preventing dust/coal ingress in equipment.

- viii. During Walk down checks of Wagon Tippler Complex chute leakages/Spillages should be noted and attended to, in priority, to avoid coal accumulation in underground spaces.
- ix. Material Collected from Suspended Magnets should be disposed regularly at designated spots to avoid accumulation/blocking walkway.
- x. Ventilation System shall be kept in service.

4.6.5.1.2. Track Hopper Complex----- Responsibility CHP dept

- i. Track Area & adjacent grizzly should be well illuminated and clear of all coal/material/debris/scrap.
- ii. Proper Arrangement of Stone handling equipment to be ensured so as to ensure safe handling of boulder/stone pieces by housekeeping staff.
- iii. There should be Lights on Paddle feeders illuminating Hydraulic system areas and operating panel areas.
- iv. Rail for Paddle Feeder Movement should be clear of coal/scrap.
- v. Availability of sump/slurry Pumps shall be ensured for underground portion of Track Hopper complex to avoid water logging/ coal slurry accumulations.
- vi. Coal collected on Paddle feeder platform must be cleaned regularly after proper precautions as per JSA.
- vii. The Drive Area/ Return side of Evacuation Conveyor should be clean. Proper Covers/Guards should be in place for preventing dust/coal ingress in equipment.
- viii. In Hydraulically Operated Paddle Feeders, cleanliness shall be ensured to avoid dust ingress in system.
- ix. During Walk down checks of Track Hopper Complex chute leakages/Spillages should be noted and attended to, in priority, to avoid coal accumulation in underground spaces.
- x. Material Collected from Suspended Magnets should be disposed regularly to avoid accumulation/blocking walkway.
- xi. Ventilation System shall be kept in service.

4.6.5.1.3. Coal Crushing Zone/ Coal Crusher House----Responsibility CHP dept

- i. All floors of Crusher House shall be cleaned regularly so that it is made clear of coal/debris/scrap/oil etc.
- ii. Any leakage/ Spillage from Coal Chutes, Vibratory Screen Feeder, Crushers, Belt Feeders shall be attended on priority.
- iii. All drive areas must be clear of coal/oil/slurry etc
- iv. Tramp Iron Chutes shall be operational and area for tramp iron opening should be suitably barricaded.

- v. Roof/Floor Drains shall be clean and drain pipes clear of any coal/scrap/debris accumulation.
- vi. Flap Gates platforms shall be cleaned regularly.
- vii. Proper Sealing of Inspection doors/Belt Feeder skirts shall be ensured to reduce dust emission in coal crusher house.
- viii. Special attention may be provided for Housekeeping activities of Coal Sampling Unit Equipment & Area

4.6.5.1.4. Coal Stack Yard ----- Responsibility CHP dept

- i. Yard Drains/Peripheral Drains shall be cleaned at regular intervals, especially before monsoon.
- ii. Retaining Walls wherever possible/feasible shall be in place so that coal sliding from piles to yard conveyor/rail may be avoided. In stations where retaining wall were not envisaged in original scheme, the same may be constructed by station.
- iii. Walkway along yard conveyors shall be clear of coal/slurry/scrap/debris etc for safe movement of men a& material.
- iv. Idler Frames/Brackets/Rollers/Deck Plates/Conveyor Return side/cable tray along yard conveyors should be clear from coal accumulation.
- v. Rail Line for long Travel of Stacker/Reclaimer should be clear from coal/slurry/scrap/debris etc.
- vi. Yard Equipment like Stacker Reclaimer should be cleaned at regular intervals and approaches shall be clear for preventive maintenance of parts of equipment.
- vii. Stock Pile Dust Suppression (SPDS) if available in station shall be in service.
- viii. Slew Platform of Stacker Reclaimer shall be clear from coal/scrap/debris etc.
- ix. Coal Pile area shall be regularly inspected for removal of scrap etc.

4.6.5.1.5. Bunker/Tripper Floor-----Responsibility CHP dept

- i. Travelling Tripper Floor shall be made clear of coal/scrap/debris/Coal dust etc.
- ii. Bunker Sealing Belt (if envisaged in original scheme) shall be available.
- iii. Conveyor Belt Drive area should be clear of coal/oil/grease/dust etc.
- iv. Idler Frames/Brackets/Rollers/Deck Plates/Conveyor Return side/cable tray along yard conveyors should be clear from coal accumulation.
- v. Proper Sealing of Inspection doors/Belt Conveyor skirts shall be ensured to reduce dust emission in Bunker Floor.
- vi. Bunker Ventilation Fans shall be kept in service.
- vii. Chute leakages/Spillages should be noted and attended to, in priority, to avoid coal accumulation in Tripper Floor.

- viii. Flap Gates platforms shall be cleaned regularly.
- ix. Manual or motorized Floor cleaning machines like ROOTS Flipper + may be adopted for safe, faster cleaning of floor areas.

4.6.5.1.6. Dozer Shed/Heavy Vehicle Shed----- Responsibility CHP dept

- i. Dozer Shed area shall be free from oil/grease/lubricant spillage.
- ii. All oil pumps/centrifuge areas must have oil spillage collection trays.
- iii. Proper arrangements/system shall be made for disposal of waste/used lubricant/oil etc. generated from dozer shed.
- iv. Adequate compliance of 5S techniques to be ensured for suitable arrangement of tools/parts etc.
- v. Storage area of/nearby diesel shed should be clean and free from coal dust/dry vegetation etc.
- vi. Any accidental spillage of fuel/lubricant/oils etc. should be adequately cleaned.

4.6.5.1.7. Conveyor Belt/Transfer Point/Conveyor Galleries--Responsibility CHP dept.

- i. Drive area of Conveyor Belts should be clear of oil/grease/dust/coal etc.
- ii. Chute leakages/Spillages should be noted and attended to, in priority, to avoid coal accumulation in transfer points.
- iii. Idler Frames/Brackets/Rollers/Deck Plates/Conveyor Return side/ cable tray along conveyors should be clear from coal accumulation.
- iv. Proper Sealing of Inspection doors/Belt Conveyor skirts shall be ensured to reduce dust emission in Transfer Points.
- v. Take Up platforms/ area adjacent to counterweight sand pit etc. shall be periodically cleaned to avoid coal accumulation.
- vi. Flap Gates platforms shall be cleaned regularly.
- vii. Tramp Iron Chutes in Transfer Points shall be operational and area for tramp iron opening should be suitably barricaded.
- viii. Roof/Floor Drains of Transfer Points shall be clean and drain pipes clear of any coal/scrap/debris accumulation.
- ix. Effectiveness of Conveyor Belt Scrappers must be ensured to reduce carryover and coal accumulation along return belt of conveyor.
- x. Rubber Curtains may be provided near discharge hoods to reduce dust outflow from discharge chutes.
- xi. Walkways should be clear from coal/slurry/scrap etc. to enable free movement of men and material in conveyor galleries.
- xii. Peripheral drains along transfer points shall be periodically cleaned.
- xiii. Coal Settling pits along conveyor galleries shall be regularly cleaned.
- xiv. All interconnecting Drains/peripheral drains from conveyor galleries/ transfer points/ Yards etc. shall not be choked and shall discharge adequately at coal slurry settling pond.

4.6.5.1.8. Salient Points----- Responsibility all depts

- i. Magnetic Plates may be procured by station to temporarily curb chute leakages and reduce spillage.
- ii. Proper Permit to work shall be taken before start of any housekeeping job.
- iii. While housekeeping of equipment requiring Electrical isolation as per JSA, proper electrical isolation shall be ensured.
- iv. Requisite Care should be taken while performing housekeeping jobs in MCC/Control Rooms.
- v. For Housekeeping of take up area/take up platforms/areas at height exposed to open areas below, only experienced personnel who have cleared vertigo test, shall be deployed.
- vi. Proper and safe approaches shall be provided for housekeeping personnel.
- vii. Staircase shall be clear of coal/stone/algae growth/slurry to avoid fouling/fall from slippery surfaces.
- viii. For carrying out housekeeping jobs, hazard identification & risk analysis and job safety analysis must be performed for enhancing safety.
- ix. Regular Toolbox Talks /safety pep talks shall be conducted for improving safety and job awareness among housekeeping manpower.
- x. Adequate Barricading wherever required shall be ensured during housekeeping jobs.
- xi. Dust Suppression shall be kept in service as per system requirement.
- xii. Drains should be clean.
- xiii. Maintenance Bay beneath Electrical Hoist shall be clear of scarp/coal/Debris so that the passage is clear for movement of men and machines for maintenance activities.
- xiv. Care shall be taken for requisite cleaning of Hoist and Accessories (including local control panels) for safe material handling.
- xv. Do's and Don'ts shall be displayed properly at workplaces and shall be regularly discussed in pep talks.
- xvi. In case of chute choking and spillages, before removal of stuck coal, first get local inspection done and then PTW shall be taken to remove coal. In no case the housekeeping personnel should stand/step/sit on conveyor belt/pulleys/deck sheet/seal plate.
- xvii. Before start of any housekeeping work, all risks/hazards shall be assessed, and requisite mitigating measures should be in place. Strict adherence to established systems of PTW/JSA/Toolbox Talk (TBT) etc. shall be done.
- xviii. It should be ensured that the personnel are properly trained and are aware of the risks involved and take all risk mitigation measures without fail.

- xix. It should be tried to remove the coal in chute/pulleys through water jet in manhole (with proper and stable platform). If required, removal of skirt rubber/boards may be done after suitable PTW. Personnel can work over belt only in case the Belt has been properly locked mechanically.
- xx. Station should develop its work instructions for regular housekeeping activities based on site conditions.
- xxi. OIN-OPS-SYST-059 Titled Guidelines on Summer and Monsoon Preparation for Coal and Gas Stations w.r.t Housekeeping for Coal Handling Plant area should be complied.
- xxii. OGN-OPS-Syst-013 Titled Manual on Handling & Storage of Coal and its respective appendix for housekeeping should be complied.
- xxiii. Sample Housekeeping Checklist is attached in Appendix for reference. Station may develop their own checklist based on specific site layout and conditions.

4.6.6. Cleaning surface Oil spillage

- i. **Absorb the Oil Stain:** Sprinkle sawdust, coconut husks, cotton jute, or a commercial oil-absorbing product on the stain. If you have none of these products on hand, set paper towels on the floor to soak up any wet oil.
- ii. If the oil spillage has taken place on mud, simply put sufficient quantity of sand over it to absorb the oil.
- iii. Allow the absorbent materials to work on the oil for 8 hours. Then, sweep up the debris with a broom and dustpan.
- iv. Repeat with fresh absorbent materials as needed to remove any remaining oil.
- v. If the oil stain remains, wash with water and soap.

4.6.7. Ash Handling Plant

The following areas should be cleaned

- i. Bottom ash area
- ii. ESP floor "O" meter area
- iii. Buffer hopper & vacuum pump area
- iv. Eco hopper & APH hopper area
- v. TAC, IAC area
- vi. Ash water pump house & Ash slurry pump house
- vii. Areas nearby ash conveying lines
- viii. Silo area
- ix. Any other area covered in ash handling system.

4.7. WASTE DISPOSAL

Waste Disposal and handling to be done as per Guidelines issued by Corporate Environment Management Group-“Zero deviation Document on Waste management”.

4.8. HOUSE KEEPING FREQUENCY

For Coal Handling Plant area, station may decide frequency of coal conveyor, transfer point, equipment, complex etc. housekeeping as per CHP conditions while maintaining clean, safe and healthy working environment

4.9. REFERENCES

- i. NFPA guidelines
- ii. OSHA guidelines
- iii. NGT report on NLC fire incident

5.0 RECORDS TO BE MAINTAINED

Sl No.	Record Description	Maintained by	Frequency	Remarks
1.	Housekeeping Records	Operation	Weekly	
2.	Housekeeping Checklist filled	Operation	weekly	
3	Housekeeping of CERs, DCS & UPS rooms, Server rooms	C&I / Operation	Daily	

6.0 ASSOCIATED DOCUMENTS

Annexure – I: Pre House-Keeping Checklist

Annexure – II: Post House-Keeping Checklist

Annexure – III: CHP Housekeeping sample Checklist

Annexure – I: Pre House-Keeping Checklist

S.No	HAZARD IDENTIFICATION	
1	Any coal dust accumulation	
2	Any ash accumulation	
3	Any coal dust leakage point identified	
4	Any ash leakage point identified.	
5	Any oil spill area	
6	Any oil leakage point.	
7	Any hot air leakage point.	
8	Presence of any material which can catch fire	
9	Any waterlogging in the area	
10	Inaccessible places if any.	
11	Walkways clear and available	
12	Spark hazard if any for coal dust cleaning	
13	Any electrical hazard, like naked wire, spark etc.	
14	Area properly lit	
15	Area to be cleaned qualifies for confined space?	
16	Proper exit path available	
17	Any hanging or loose object over head?	
18	Firm surface available for standing and cleaning.	
19	Source of leakage or waste material to be identified and shared with the concerned CCR for raising WOC.	
	HOUSE KEEPING METHODOLOGY TO BE ADOPTED	
19	Which method is used for housekeeping? 1. Water washing 2. Vacuum cleaning 3. Air blowing (by compressed air or handheld blower) 4. Brooming.	
20	If Vacuum cleaning is to be used, Grounding facility available Flame proof approved vacuum cleaner available	
21	If water washing is to be used Water source available Hose and nozzle available	
22	If brooming method is used Sufficient number of brooms available	
23	If Air blowing method is used Compressed air source available Compressed air hose available Compressed air nozzle available	
	PPE TO BE USED	
24	Gloves	yes
25	Mask	yes
26	Earmuff	
27	Safety shoes	yes
28	Goggles	yes
29	Flame resistant clothing	

Annexure – I: Post House-Keeping Checklist

S.No.	Description	Y(Yes) / N(No)/ Remarks
1	Debris is cleared from work surfaces, passages, and stairs.	
2	Storage areas and walkways are reasonably free of dangerous depressions, holes, obstructions, and debris.	
3	All walking and working surfaces are reasonably dry and free from grease or oil.	
4	Spills of oil, grease, and other liquids are removed or cleaned.	
5	Waste or trash containers are emptied.	
6	If any oil leakage persists proper capturing system available	
7	All walking and working surfaces are free of protruding nails.	
8	Nails are bent down or removed before scrap material is discarded.	
9	Any object protruding at head height has been removed or flagged.	
10	All tools and gears used for housekeeping are cleared	
11	Source of leakage or waste material to be identified and shared with the concerned CCR for raising WOC.	
	HAZARDOUS MATERIAL STORAGE AND DISPOSAL	
12	Flammable and combustible scrap, debris, and waste are removed from buildings or structures.	
13	Appropriate cleanup materials are available for leaks or spills of flammables or other hazardous materials.	
	SANITATION	
14	Toilets and washing facilities are clean and sanitary.	
	ENVIRONMENT	
15	Lighting and ventilation are adequate.	
16	Burned out lights are reported and replaced.	

Annexure – I: CHP House-Keeping sample Checklist
COAL HANDLING PLANT-Housekeeping Sample Checklist

NOTE: Cleaning of Deck plate, below return belt and all walkways to be checked for all conveyors and no jamming of idler and pulley to be ensured.					Date:
Sr. No.	Area	Cleaning Status	Scrap	Coal Leakage(Chute Leak/Skirt Damage etc.)	Remarks
1	TP-10, 14A/B Head End Floor				
2	TP-10, Zero & Middle Floor				
3	TP-15, Tripper Head End Floor				
4	TP-15, Zero & Middle Floor				
5	Conveyor 19A/B				
7	TP-17 19A/B Head End Floor				
8	TP-17 Zero & Middle Floor				
9	TP-17 Take-up Floor				
10	Conveyor 28				
12	TP-24 Conv. 28 Head end Floor				
13	TP-24 Zero & middle floor				
14	Conveyor 20				
15	Conv. 20 Drainage				
16	Stacker/Reclaimer - 2				
17	Conv. 29 Drainage				

Note: Above checklist is just an illustration. Station may prepare own checklist based on station layout and requirements/conditions.